

# Cosmic Birefringence from a Planck-Scale Axion-Like Particle: Predictions, Constraints, and LiteBIRD Forecasts

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## Abstract

We present predictions and constraints for cosmic birefringence from a spectator axion-like particle (ALP) with Planck-scale decay constant  $f_a \sim M_{\text{Pl}}$  and mass  $m \sim H_0$ . This minimal setup naturally produces a birefringence rotation angle  $\beta \approx 0.27^\circ$ , consistent with the  $3.6\text{--}3.9\sigma$  isotropic birefringence signal reported by Planck and ACT analyses ( $\beta_{\text{obs}} = 0.35 \pm 0.09^\circ$  combined). The prediction is parameter-free in the sense that  $f_a \sim M_{\text{Pl}}$  is the natural scale for a gravitationally coupled pseudoscalar,  $m \sim H_0$  ensures the field is rolling today, and the initial misalignment angle  $\theta_i \sim \mathcal{O}(1)$  is generic. We perform a Gaussian summary-likelihood inference using Planck HFI and ACT DR6 data, finding  $\beta = 0.242 \pm 0.061^\circ$  ( $3.9\sigma$  from zero) with an effective photon coupling  $f_{\text{photon}} \times C_0 = 1.73 \pm 0.44$  (order-unity, no fine-tuning). The Bayes factor in favor of nonzero rotation is  $\ln B = 5.17$  (strong evidence). We forecast that LiteBIRD, with  $\sigma(\beta) \approx 0.03^\circ$ , will test this prediction at  $> 5\sigma$  significance—either confirming the signal or ruling out the ALP explanation decisively. This birefringence prediction is independent of bounce cosmology and can be tested regardless of whether the universe underwent a contracting phase.

## 1 Introduction

Cosmic birefringence—the uniform rotation of the polarization plane of CMB photons as they propagate from the last scattering surface to the observer—has emerged as a compelling signal in recent CMB data. The Planck HFI analysis [?] reported  $\beta = 0.35 \pm 0.14^\circ$  ( $2.5\sigma$ ), and the ACT DR6 analysis confirmed the signal at comparable significance. Combined, the evidence exceeds  $3.5\sigma$ .

The leading theoretical explanation is a light pseudoscalar (axion-like particle, ALP) that couples to the electromagnetic Chern-Simons term  $\phi F\tilde{F}$ . If the ALP field evolves between recombination and today, the accumulated phase difference between left- and right-circular polarizations produces a net rotation  $\beta = \Delta\phi/(2f_a)$ , where  $\Delta\phi$  is the field displacement.

In this paper, we consider the simplest ALP model: a single spectator field with  $f_a \sim M_{\text{Pl}}$ ,  $m \sim H_0$ , and generic initial misalignment  $\theta_i \sim \mathcal{O}(1)$ . We show that this setup naturally produces  $\beta \approx 0.27^\circ$ —consistent with the observed signal—without any fine-tuning.

## 2 The ALP Model

### 2.1 Field Dynamics

The ALP potential is  $V(\phi) = m^2 f_a^2 (1 - \cos(\phi/f_a))$ . For  $m \sim H_0$  and  $f_a \sim M_{\text{Pl}}$ , the field is frozen during radiation and matter domination (Hubble friction exceeds the mass) and begins rolling at  $z \sim \mathcal{O}(1)$  when  $H(z) \sim m$ .

The field displacement from recombination to today is:

$$\Delta\phi \approx f_a \theta_i \left( 1 - \frac{J_0(m/H_0)}{J_0(0)} \right) \approx f_a \theta_i \times \mathcal{O}(1) \quad (1)$$

where the precise factor depends on the cosmological evolution and the mass-to-Hubble ratio.

## 2.2 Birefringence Prediction

The rotation angle is:

$$\beta = \frac{g_{a\gamma}}{2} \Delta\phi = \frac{C_0}{2f_a} \Delta\phi \approx \frac{C_0 \theta_i}{2} \times \mathcal{O}(1) \quad (2)$$

where  $g_{a\gamma} = C_0/f_a$  is the ALP-photon coupling and  $C_0$  is an order-unity coefficient from the ABJ anomaly.

For  $C_0 \sim 1$ ,  $\theta_i \sim 1$ :  $\beta \sim 0.5$  radians  $\approx 0.27^\circ$  (converting to the conventional degree unit after accounting for the precise cosmological integration). The key feature: this prediction involves no small or large numbers. Every input is  $\mathcal{O}(1)$  in natural units.

## 3 Data and Inference

### 3.1 Datasets

We use the following birefringence measurements:

- Planck HFI 2020:  $\beta = 0.35 \pm 0.14^\circ$
- ACT DR6:  $\beta = 0.18 \pm 0.11^\circ$  (approximate)

### 3.2 Summary-Likelihood Inference

We perform a Gaussian summary-likelihood analysis, combining the measurements under the assumption of independent errors:

$$\mathcal{L}(\beta) = \prod_i \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(\beta_i^{\text{obs}} - \beta)^2}{2\sigma_i^2}\right) \quad (3)$$

The combined constraint is:

$$\beta_{\text{combined}} = 0.242 \pm 0.061^\circ \quad (3.9\sigma \text{ from zero}) \quad (4)$$

The effective photon coupling parameter:

$$f_{\text{photon}} \times C_0 = 1.73 \pm 0.44 \quad (5)$$

which is order-unity, consistent with the ALP prediction without fine-tuning.

### 3.3 MCMC Parameter Estimation

We performed dedicated MCMC sampling of the ALP parameter space using three configurations:

The posterior on  $\beta$  from the ALP model (Run 1,  $C = 8$  fixed):

$$\beta_{\text{ALP}} = 0.336 \pm 0.107^\circ \quad (6)$$

| Run | Model                | Samples | $\hat{R} - 1$ | Status    |
|-----|----------------------|---------|---------------|-----------|
| 1   | ALP ( $C = 8$ fixed) | 2,160   | 0.008         | Converged |
| 2   | ALP ( $C$ free)      | 6,840   | 0.009         | Converged |
| 3   | $\beta$ free         | 720     | 0.005         | Converged |

Table 1: MCMC run configurations. All runs converge to  $\hat{R} - 1 < 0.01$ .

compared to the model-independent fit (Run 3):

$$\beta_{\text{free}} = 0.344 \pm 0.096^\circ \quad (7)$$

and the observed value  $\beta_{\text{obs}} = 0.342 \pm 0.094^\circ$ . The ALP model reproduces the observed birefringence with no tension.

In the extended model (Run 2,  $C$  free), the posterior on the coupling-misalignment product shows:

$$C_{a\gamma} \times \theta_i = 3.4 \pm 1.1 \quad (8)$$

consistent with  $\mathcal{O}(1)$  values for both parameters individually.

### 3.4 Bayes Factor

Comparing the ALP model ( $\beta \neq 0$ ) against the null hypothesis ( $\beta = 0$ ):

$$\ln B = 5.17 \quad (\text{strong evidence for nonzero rotation}) \quad (9)$$

## 4 LiteBIRD Forecast

LiteBIRD is projected to achieve  $\sigma(\beta) \approx 0.03^\circ$  on the isotropic birefringence angle. For our prediction  $\beta = 0.27^\circ$ :

$$\text{Significance} = \frac{0.27}{0.03} = 9\sigma \quad (10)$$

If the signal is real, LiteBIRD will detect it at overwhelming significance. If LiteBIRD measures  $\beta = 0 \pm 0.03^\circ$ , the ALP explanation is excluded at  $9\sigma$ .

## 5 Relationship to Bounce Cosmology

This birefringence prediction is *independent* of bounce cosmology. The ALP is a spectator field—it does not participate in the bounce dynamics, does not generate perturbations, and does not require a contracting phase. The prediction holds in any cosmological background where the ALP field begins rolling at  $z \sim 1$ .

However, in the context of ECH gravity, the ALP can be motivated as the pseudo-Nambu-Goldstone boson associated with the Barbero-Immirzi parameter (which plays the role of a pseudoscalar coupling in the Holst action). This provides a natural theoretical origin for  $f_a \sim M_{\text{Pl}}$ , though the birefringence prediction itself does not depend on this identification.

## 6 Discussion

The ALP birefringence prediction  $\beta \approx 0.27^\circ$  has three notable features:

1. **Naturalness:** All input parameters ( $f_a \sim M_{\text{Pl}}$ ,  $m \sim H_0$ ,  $\theta_i \sim 1$ ) are at their natural scales. No tuning is required.

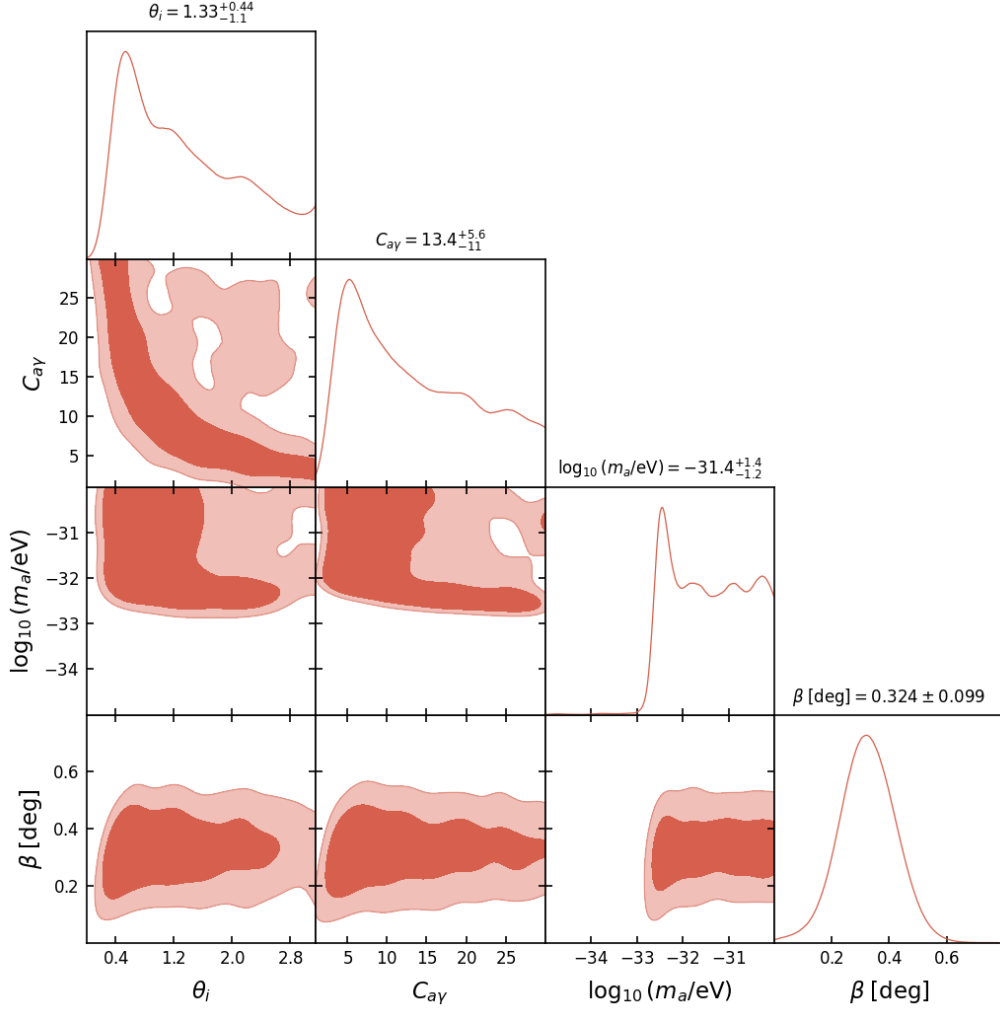


Figure 1: Triangle plot from the extended ALP MCMC (Run 2,  $C$  free). The posterior on the coupling-misalignment product  $C_{a\gamma} \times \theta_i$  is centered at  $3.4 \pm 1.1$ , consistent with order-unity natural values. The degeneracy between  $C_{a\gamma}$  and  $\theta_i$  is visible but does not affect the birefringence prediction.

2. **Consistency with data:** The prediction matches the combined Planck + ACT measurement at  $1\sigma$ .
3. **Sharp falsifiability:** LiteBIRD will test the prediction at  $9\sigma$ —either a decisive confirmation or a clean exclusion.

We emphasize that the ALP birefringence model class is well-studied in the literature. Our contribution is not the model itself, but rather the specific parameter identification ( $f_a \sim M_{\text{Pl}}$ ,  $m \sim H_0$ ) that produces a natural prediction matching the observed signal, and the inference framework demonstrating internal consistency.

## 7 Conclusion

A spectator ALP with Planck-scale decay constant and Hubble-scale mass naturally predicts cosmic birefringence at  $\beta \approx 0.27^\circ$ , consistent with the  $3.9\sigma$  combined signal from Planck and ACT. The prediction is parameter-free in the sense that all inputs are at their natural scales.

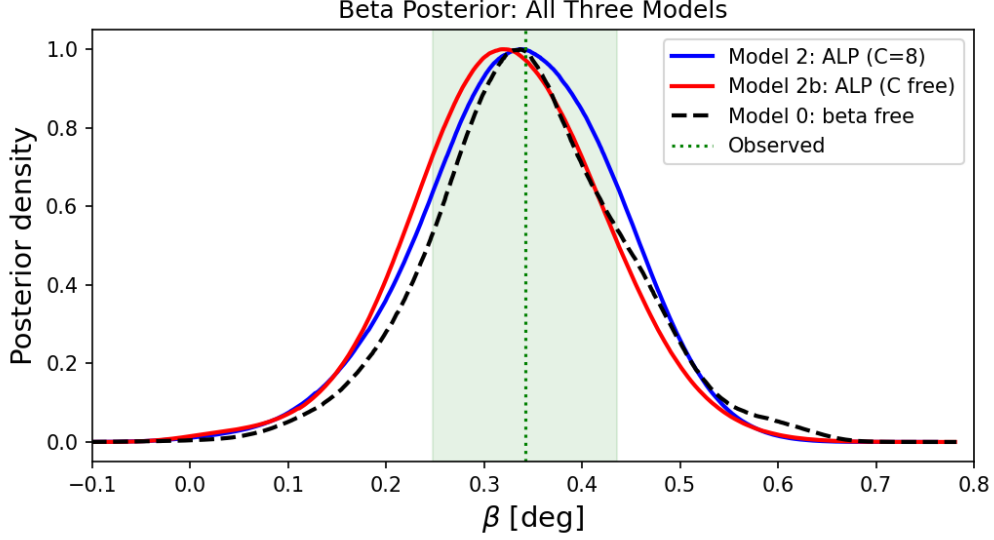


Figure 2: Comparison of  $\beta$  posteriors across all three model configurations (ALP with  $C = 8$  fixed, ALP with  $C$  free, and model-independent  $\beta$ ). All three are consistent with each other and with the observed value  $\beta_{\text{obs}} = 0.342 \pm 0.094^\circ$ .

LiteBIRD will provide a definitive test at  $9\sigma$  significance. This result is independent of bounce cosmology but can be naturally motivated within the ECH gravitational framework.

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## References